Chapter III
De Gravitatione [date unknown]

It is fitting to treat the science of the weight and of the equilibrium of fluids and solids in fluids by a twofold method. To the extent that it appertains to the mathematical sciences, it is reasonable that I largely abstract it from physical considerations. And for this reason I have undertaken to demonstrate its individual propositions from abstract principles, sufficiently well known to the student, strictly and geometrically. Since this doctrine may be judged to be somewhat akin to natural philosophy, in so far as it may be applied to making clear many of the phenomena of natural philosophy and in order, moreover, that its usefulness may be particularly apparent and the certainty of its principles perhaps confirmed, I shall not be reluctant to illustrate the propositions abundantly from experiments as well, in such a way, however, that this freer method of discussion, disposed in scholia, may not be confused with the former, which is treated in lemmas, propositions and corollaries.

The foundations from which this science may be demonstrated are either definitions of certain words, or axioms and postulates no one denies. And of these I treat directly.

Definitions

The terms 'quantity', 'duration', and 'space' are too well known to be susceptible of definition by other words.[1]

[1] Cf. Newton's discussion in the Scholium to the *Principia* in this volume (p. 84).

Definition I. Place is a part of space which something fills completely.
Definition 2. Body is that which fills place.
Definition 3. Rest is remaining in the same place.
Definition 4. Motion is change of place.

Note. I said that a body fills place, that is, it so completely fills it that it wholly excludes other things of the same kind or other bodies, as if it were an impenetrable being. Place could be said, however, to be a part of space into which a thing enters completely; but as only bodies are here considered and not penetrable things, I have preferred to define [place] as the part of space that a thing fills.

Moreover, since body is here proposed for investigation not in so far as it is a physical substance endowed with sensible qualities, but only in so far as it is extended, mobile, and impenetrable, I have not defined it in a philosophical manner, but abstracting the sensible qualities (which philosophers also should abstract, unless I am mistaken, and assign to the mind as various ways of thinking excited by the motions of bodies),[2] I have postulated only the properties required for local motion. So that instead of physical bodies you may understand abstract figures in the same way that they are considered by geometers when they assign motion to them, as is done in Euclid's *Elements*, Book I, 4 and 8. And in the demonstration of the tenth definition, Book XI, this should be done, since it is mistakenly included among the definitions and ought rather to be demonstrated among the propositions, unless perhaps it should be taken as an axiom.[3]

Moreover, I have defined motion as change of place because motion, transition, translation, migration, and so forth seem to be synonymous words. If you prefer, let motion be transition or translation of a body from place to place.

[2] Newton refers here to the distinction between what came to be known as primary and secondary qualities, a distinction first articulated, in the modern period, by Galileo in his *Assayer* and expanded on by Boyle and Locke, among others.

[3] In Book I, proposition 4, Euclid's proof of the congruence of two triangles involves the motion of one triangle such that it achieves superposition with the other; proposition 8 similarly employs the so-called method of superposition. Definition 10 of Book XI reads as follows: "Equal and similar solid figures are those contained by similar planes equal in multitude and in magnitude" (*The Thirteen Books of Euclid's Elements*, ed. and trans. Thomas Heath (Cambridge University Press, 1926), vol. III, 261). Newton takes the familiar position that this is properly understood to be a theorem rather than a definition. Some take it to be demonstrable as a theorem through the method of superposition, which may be why Newton mentions it in tandem with the above propositions from Book I.

For the rest, when I suppose in these definitions that space is distinct from body, and when I determine that motion is with respect to the parts of that space, and not with respect to the position of neighbouring bodies, lest this should be taken as being gratuitously contrary to the Cartesians, I shall venture to dispose of his fictions.

I can summarize his doctrine in the following three propositions:

(1) That from the truth of things only one proper motion[4] fits each body (*Principles*, Part II, articles 28, 31, 32), which may be defined as being the translation of one part of matter or of one body from the vicinity of those bodies that immediately adjoin it, and which are regarded as being at rest, to the vicinity of others (*Principles*, Part II, article 25; Part III, article 28).[5]

(2) That by a body – transferred in its proper motion according to this definition – may be understood not only some particle of matter, or a body composed of parts relatively at rest, but all that is transferred simultaneously, although this may, of course, consist of many parts which have different relative motions (*Principles*, Part II, article 25).

(3) That besides this motion particular to each body there can arise in it innumerable other motions through participation (or in so far as it is part of other bodies having other motions) (*Principles*, Part II, article 31), which, however, are not motions in the philosophical sense and rationally speaking (Part III, article 29) and according to the truth of things (Part II, article 25 and Part III, article 28), but only improperly and according to common sense (Part II, articles 24, 25, 28, 31; Part III, article 29). That kind of motion he seems to describe (Part II, article 24; Part III, article 28) as the action by which any body migrates from one place to another.

[4] Newton refers here, and elsewhere, to Descartes's distinction in the *Principles* between the "common" (literally, "vulgar" or "loose") and the "proper" understanding of motion (Part II, articles 34–5); cf. Newton's own distinction between "mathematical" and "common" conceptions of space, time, and motion in the Scholium to the *Principia* (pp. 84–91 below).

[5] In these sections of Part II of his *Principles*, Descartes defines and discusses motion, continuing on to present his laws of nature, where he articulates, among other things, an early version of the principle of inertia. In the sections of Part III cited by Newton, Descartes claims that, properly speaking, the earth does not move, given his earlier definition of motion in the *Principles*.

And just as he formulates two types of motion, namely proper and derivative, so he assigns two types of place from which these motions proceed, and these are the surfaces of immediately surrounding bodies (Part II, article 15), and the position among any other bodies (Part II, article 13; Part III, article 29).

Indeed, not only do its absurd consequences convince us how confused and incongruous with reason this doctrine is, but Descartes seems to acknowledge the fact by contradicting himself. For he says that speaking properly and according to philosophical sense, the earth and the other planets do not move, and that he who claims they are moved because of their translation with respect to the fixed stars speaks without reason and only in the common fashion (Part III, articles 26, 27, 28, 29). Yet later he attributes to the earth and planets a tendency to recede from the sun as though from a centre about which they are moved circularly, by which they are balanced at their own distances from the sun by a similar tendency of the gyrating vortex (Part III, article 140). What, then? Is this tendency to be derived from the (according to Descartes) true and philosophical rest of the planets, or rather from [their] common and non-philosophical motion? But Descartes says further that a comet has a lesser tendency to recede from the sun when it first enters the vortex, and maintaining a position among the fixed stars does not yet obey the impetus of the vortex, but with respect to it is transferred from the vicinity of the contiguous aether and so philosophically speaking gyrates round the sun, after which the matter of the vortex carries the comet along with it and so renders it at rest, according to strict philosophical sense (Part III, articles 119–20). And so the philosopher is hardly consistent who uses as the basis of philosophy the common motion which he had rejected a little before, and now rejects that motion as fit for nothing which alone was formerly said to be true and philosophical, according to the nature of things. And since the gyrating of the comet around the sun in his philosophical sense does not cause a tendency to recede from the centre, which a gyration in the common sense can do, surely motion ought to be acknowledged in the common sense, rather than the philosophical.

Secondly, he seems to contradict himself when he postulates that to each body corresponds a strict motion, according to the nature of things; and yet he asserts that motion to be a product of our imagination, defining it as translation from the vicinity of bodies which are not at rest but only are seen to be at rest, even though they may instead be moving, as is more fully explained in Part II, articles 29–30. And by this he aims to avoid the difficulties concerning the mutual translation of bodies, namely, why one body is said to move rather than another, and why a boat on a flowing stream is said to be at rest when it does not change its position with respect to the banks (Part II, article 15). But so that the contradiction may be evident, imagine that someone sees the matter of the vortex to be at rest, and that the earth, philosophically speaking, is at rest at the same time; imagine also that at the same time someone else sees that the same matter of the vortex is moving in a circle, and that the earth, philosophically speaking, is not at rest. In the same way, a ship at sea will simultaneously move and not move; and that is so without taking motion in the looser common sense, according to which there are innumerable motions for each body, but in his philosophical sense, according to which, he says, there is but one in each body, and that one proper to it and corresponding to the nature of things and not to our imagination.

Thirdly, he seems hardly consistent when he posits a single motion that corresponds to each body according to the truth of things, and yet (Part II, article 31) posits innumerable motions that really are in each body. For the motions that really are in any body are in fact natural motions, and thus [are] motions in the philosophical sense and according to the truth of things, even though he would contend that they are motions in the common sense only. Add that when a whole thing moves, all the parts that constitute the whole and are translated together are really at rest, unless it is truly conceded that they move by participating in the motion of the whole, and then indeed they have innumerable motions according to the truth of things.

But besides this, we may see from its consequences how absurd this doctrine of Descartes is. And first, just as he pointedly contends that the earth does not move because it is not transferred from the vicinity of the contiguous aether, so from these very same principles it would follow that the internal particles of hard bodies, while they are not transferred from the vicinity of immediately contiguous particles, do not have motion in the strict sense, but move only by participating in the motion of the external particles. It rather appears that the interior parts of the external particles do not move with a proper motion because they are not transferred from the vicinity of the internal parts, and I submit that only the external surface of each body moves with a proper motion and that the whole internal substance, that is the whole body, moves through participation in the motion of the external surface. The fundamental definition of motion errs, therefore, that attributes to bodies that which is suitable only to surfaces, and which denies that there can have been a more proper motion of any body at all.

Secondly, if we regard only article 25 of Part II, each body has not merely a single proper motion but innumerable ones, provided that they are said to be moved properly and according to the truth of things by which the whole is properly moved. And that is because by the body whose motion he defines, he understands all that which is transferred simultaneously, and yet this may consist of parts having other motions among themselves: [for example] a vortex together with all the planets, or a ship along with everything in it floating on the sea, or a man walking on a ship together with the things he carries with him, or the wheel[s] of a clock together with its constituent metallic particles. For unless you say that the motion of the whole aggregate is not posited as proper motion and as belonging to the parts according to the truth of things, it will have to be admitted that all these motions of the wheels of the clock, of the man, of the ship, and of the vortex, are truly and philosophically speaking in the particles of the wheels [of the man, of the ship, and of the vortex].

From both of these consequences it appears further that no one motion can be said to be true, absolute and proper in preference to others, but that all – whether with respect to contiguous bodies or remote ones – are equally philosophical; and nothing more absurd than that can be imagined. For unless it is conceded that there can be a single physical motion of any body, and that the rest of its changes of relation and position with respect to other bodies are just external designations, it follows that the earth (for example) endeavours to recede[6] from the centre of the sun on account of a motion relative to the fixed stars, and endeavours the less to recede on account of a lesser motion relative to Saturn and the aetherial orbit in which it is carried, and still less relative

[6] We have translated Newton's "conatus" throughout as "endeavour" for two reasons. First, when writing in English and expressing related points, Newton himself uses "endeavour"; see, for instance, the letter to Boyle (this volume, p. 18). Second, Cohen translates it in this way in his "Guide to the *Principia*" (pp. 14–15), which is prefixed to the new standard translation of that work, from which we have reprinted excerpts here – see Note on texts and translations above.

to Jupiter and the swirling aether which occasions its orbit, and also less relative to Mars and its aetherial orbit, and much less relative to other orbits of aetherial matter which, although not bearing planets, are closer to the annual orbit of the earth; and indeed relative to its own orbit it has no endeavour, because it does not move in it. Since all these endeavours and non-endeavours cannot absolutely coincide, it is rather to be said that only the natural and the absolute motion of the earth coincide, on account of which it endeavours to recede from the sun, and because of which its translations with respect to external bodies are just external designations.

Thirdly, it follows from the Cartesian doctrine that motion can be generated where there is no force acting. For example, if God should suddenly cause the spinning of our vortex to stop, without applying any force to the earth which could stop it at the same time, Descartes would say that the earth is moving in a philosophical sense – on account of its translation from the vicinity of the contiguous fluid – whereas before he said it was at rest, in the same philosophical sense.

Fourthly, it also follows from the same doctrine that God himself could not generate motion in some bodies even though he impelled them with the greatest force. For example, if God impelled the starry heaven together with all the most remote part of creation with any very great force so as to cause it to revolve around the earth (suppose with a diurnal motion): yet by this, according to Descartes, the earth alone and not the sky would be truly said to move (Part III, article 38), as if it would be the same whether, with a tremendous force, he would cause the skies to turn from east to west, or with a small force turn the earth in the opposite direction. But who will suppose that the parts of the earth endeavour to recede from its centre on account only of a force impressed upon the heavens? Or is it not more agreeable to reason that when a force imparted to the heavens makes them endeavour to recede from the centre of the gyration thus caused, they are for that reason the sole bodies properly and absolutely moved; and that when a force impressed upon the earth makes its parts endeavour to recede from the centre of gyration thus caused, for that reason it is the sole body properly and absolutely moved, although there is the same relative motion of the bodies in both cases. And thus physical and absolute motion is to be designated by considerations other than translation, such translation being a merely external designation.

Fifthly, it seems repugnant to reason that bodies should change their relative distances and positions without physical motion; but Descartes says that the earth and the other planets and the fixed stars are properly speaking at rest, and nevertheless they change their relative positions.

Sixthly, on the other hand, it seems no less repugnant to reason that of several bodies maintaining the same relative positions one should move physically while others are at rest. But if God should cause any planet to stand still and make it continually maintain the same position with respect to the fixed stars, would not Descartes say that although the stars are not moving, the planet now moves physically on account of a translation from the matter of the vortex?

Seventhly, I ask by what reason any body is properly said to move when other bodies from whose vicinity it is transferred are not seen to be at rest, or rather when they cannot be seen to be at rest. For example, in what way can our own vortex be said to move circularly on account of the translation of matter near the circumference, from the vicinity of similar matter in other surrounding vortices, since the matter of surrounding vortices cannot be seen to be at rest, and this not only with respect to our vortex, but also in so far as those vortices are not at rest with respect to each other. For if the philosopher refers this translation not to the numerical corporeal particles of the vortices, but to the generic space (as he calls it) in which those vortices exist, at last we do agree, for he admits that motion ought to be referred to space in so far as it is distinguished from bodies.

Lastly, that the absurdity of this position may be disclosed in full measure, I say that it follows furthermore that a moving body has no determinate velocity and no definite line in which it moves. And, what is worse, that the velocity of a body moving without resistance cannot be said to be uniform, nor the line said to be straight in which its motion is accomplished. On the contrary, there can be no motion since there can be none without a certain velocity and determination.

But that this may be clear, it is first of all to be shown that when a certain motion is finished it is impossible, according to Descartes, to assign a place in which the body was at the beginning of the motion; it cannot be said from where the body moved. And the reason is that according to Descartes, the place cannot be defined or assigned except with respect to the position of the surrounding bodies, and after the completion of some motion the position of the surrounding bodies no longer stays the same as it was before. For example, if the place of the planet Jupiter a year ago were sought now, by what procedure, I ask, can the Cartesian philosopher describe it? Not by means of the positions of the particles of the fluid matter, for the positions of these particles have greatly changed since a year ago. Nor can he describe it by the positions of the sun and fixed stars, for the unequal influx of subtle matter through the poles of the vortices towards the central stars (Part III, article 104), the undulation (article 114), inflation (article 111) and absorption of the vortices, and other more true causes, such as the rotation of the sun and stars around their own centres, the generation of spots, and the passage of comets through the heavens, change both the magnitude and positions of the stars so much that they may be adequate to designate the place sought only with an error of several miles; and still less can the place be accurately described and determined by their help, as geometry would require it to be described. Truly there are no bodies in the world whose relative positions remain unchanged with the passage of time, and certainly none which do not move in the Cartesian sense: that is, which are neither transported from the vicinity of contiguous bodies, nor are parts of other bodies so translated. And thus there is no basis from which we can at the present moment designate a place which was in the past, or say that such a place is any longer discoverable in nature. For since, according to Descartes, place is nothing but the surface of surrounding bodies or position among some other more distant bodies, it is impossible (according to his doctrine) that it should exist in nature any longer than those bodies maintain the same positions from which he takes the individual designation. And so, reasoning as in the question of Jupiter's position a year ago, it is clear that if one follows Cartesian doctrine, not even God himself could define the past position of any moving body accurately and geometrically now that a fresh state of things prevails since, on account of the changed positions of the bodies, the place does not exist in nature any longer.

Now since it is impossible to pick out the place in which a motion began – that is, the beginning of the space traversed – for this place no longer exists after the motion is completed, that traversed space, having no beginning, can have no length; and since velocity depends upon the length of the space passed over in a given time, it follows that the moving body can have no velocity, just as I wished to show at first. Moreover, what was said regarding the beginning of the space passed over should be understood concerning all the intermediate places; and thus, as the space has no beginning nor intermediate parts, it follows that there was no space passed over and thus no determinate motion, which was my second point. It follows indubitably that Cartesian motion is not motion, for it has no velocity, no determination, and there is no space or distance traversed by it. So it is necessary that the definition of places, and hence of local motion, be referred to some motionless being such as extension alone or space in so far as it is seen to be truly distinct from bodies. And this the Cartesian philosopher may the more willingly allow, if only he notices that Descartes himself had an idea of extension as distinct from bodies, which he wished to distinguish from corporeal extension by calling it generic (*Principles*, Part II, articles 10, 12, 18). And also that the rotations of the vortices, from which he deduced the force of the aether in receding from their centres, and thus the whole of his mechanical philosophy, are tacitly referred to generic extension.

In addition, since Descartes in Part II, articles 4 and 11, seems to have demonstrated that body does not differ at all from extension, abstracting hardness, colour, weight, cold, heat, and the remaining qualities which body can lack, so that at last there remains only its extension in length, width, and depth, which therefore alone pertain to its essence. And as this has been taken by many as proved, and is in my view the only reason for having confidence in this opinion, and lest any doubt should remain about the nature of motion, I shall reply to this argument by saying what extension and body are, and how they differ from each other. For since the distinction of substances into thinking and extended [entities], or rather into thoughts and extensions, is the principal foundation of Cartesian philosophy, which he contends to be known more exactly than mathematical demonstrations: I consider it most important to overthrow [that philosophy] as regards extension, in order to lay truer foundations of the mechanical sciences.[7]

Perhaps now it may be expected that I should define extension as substance, or accident, or else nothing at all. But by no means, for it has its own manner of existing which is proper to it and which fits neither substances nor accidents. It is not substance: on the one hand, because it

[7] The distinction between thinking and extended substances is obviously crucial in Descartes's *Meditations*, which Newton read. At the beginning of this paragraph, Newton may have had the wax example from the Second Meditation in mind.

is not absolute in itself, but is as it were an emanative effect of God and an affection of every kind of being; on the other hand, because it is not among the proper affections that denote substance, namely actions, such as thoughts in the mind and motions in body. For although philosophers do not define substance as an entity that can act upon things, yet everyone tacitly understands this of substances, as follows from the fact that they would readily allow extension to be substance in the manner of body if only it were capable of motion and of sharing in the actions of body. And on the contrary, they would hardly allow that body is substance if it could not move, nor excite any sensation or perception in any mind whatsoever. Moreover, since we can clearly conceive extension existing without any subject, as when we may imagine spaces outside the world or places empty of any body whatsoever, and we believe [extension] to exist wherever we imagine there are no bodies, and we cannot believe that it would perish with the body if God should annihilate a body, it follows that [extension] does not exist as an accident inhering in some subject. And hence it is not an accident. And much less may it be said to be nothing, since it is something more than an accident, and approaches more nearly to the nature of substance. There is no idea of nothing, nor has nothing any properties, but we have an exceptionally clear idea of extension by abstracting the dispositions and properties of a body so that there remains only the uniform and unlimited stretching out of space in length, breadth and depth. And furthermore, many of its properties are associated with this idea; these I shall now enumerate not only to show that it is something, but also to show what it is.

1. In all directions, space can be distinguished into parts whose common boundaries we usually call surfaces; and these surfaces can be distinguished in all directions into parts whose common boundaries we usually call lines; and again these lines can be distinguished in all directions into parts which we call points. And hence surfaces do not have depth, nor lines breadth, nor points dimension, unless you say that coterminous spaces penetrate each other as far as the depth of the surface between them, namely what I have said to be the boundary of both or the common limit; and the same applies to lines and points. Furthermore, spaces are everywhere contiguous to spaces, and extension is everywhere placed next to extension, and so there are everywhere common boundaries of contiguous parts; that is, there are everywhere surfaces acting as boundaries to solids on this side and that; and everywhere lines in which

parts of the surfaces touch each other; and everywhere points in which the continuous parts of lines are joined together. And hence there are everywhere all kinds of figures, everywhere spheres, cubes, triangles, straight lines, everywhere circular, elliptical, parabolical, and all other kinds of figures, and those of all shapes and sizes, even though they are not disclosed to sight. For the delineation of any material figure is not a new production of that figure with respect to space, but only a corporeal representation of it, so that what was formerly insensible in space now appears before the senses. For thus we believe all those spaces to be spherical through which any sphere ever passes, being progressively moved from moment to moment, even though a sensible trace of that sphere no longer remains there. We firmly believe that the space was spherical before the sphere occupied it, so that it could contain the sphere; and hence as there are everywhere spaces that can adequately contain any material sphere, it is clear that space is everywhere spherical. And so of other figures. In the same way we see no material shapes in clear water, yet there are many in it which merely introducing some colour into its parts will cause to appear in many ways. However, if the colour were introduced, it would not constitute material shapes, but only cause them to be visible.

2. Space is extended infinitely in all directions. For we cannot imagine any limit anywhere without at the same time imagining that there is space beyond it. And hence all straight lines, paraboloids, hyperboloids, and all cones and cylinders and other figures of the same kind continue to infinity and are bounded nowhere, even though they are crossed here and there by lines and surfaces of all kinds extending transversely, and with them form segments of figures in all directions. So that you may indeed have an instance of infinity, imagine any triangle whose base and one side are at rest and the other side turns about the contiguous end of its base in the plane of the triangle so that the triangle is by degrees opened at the vertex, and meanwhile take a mental note of the point where the two sides meet, if they are produced that far: it is obvious that all these points are found on the straight line along which the fixed side lies, and that they become perpetually more distant as the moving side turns further until the two sides become parallel and can no longer meet anywhere. Now, I ask, what was the distance of the last point where the sides met? It was certainly greater than any assignable distance, or rather none of the points was the last,

and so the straight line in which all those meeting points lie is in fact greater than finite. Nor can anyone say that this is infinite only in imagination, and not in fact; for if a triangle is actually drawn, its sides are always, in fact, directed towards some common point, where both would meet if produced, and therefore there is always such an actual point where the produced sides would meet, although it may be imagined to fall outside the limits of the physical universe. And so the line traced by all these points will be real, though it extends beyond all distance.

If anyone now objects that we cannot imagine extension to be infinite, I agree. But at the same time I contend that we can understand it. We can imagine a greater extension, and then a greater one, but we understand that there exists a greater extension than any we can imagine. And here, incidentally, the faculty of understanding is clearly distinguished from imagination.

Should one say further that we do not understand what an infinite being is, save by negating the limitations of a finite being, and that this is a negative and faulty conception, I deny this. For the limit or boundary is the restriction or negation of greater reality or existence in the limited being, and the less we conceive any being to be constrained by limits, the more we observe something to be attributed to it, that is, the more positively we conceive it. And thus by negating all limits the conception becomes maximally positive. 'End' [*finis*] is a word negative with respect to perception, and thus 'infinity', since it is the negation of a negation (that is, of ends), will be a word maximally positive with respect to our perception and understanding, though it seems grammatically negative. Add [also] that positive and finite quantities of many surfaces infinite in length are accurately known to geometers. And so I can positively and accurately determine the solid quantities of many solids infinite in length and breadth and compare them to given finite solids. But this is irrelevant here.

If Descartes should now say that extension is not infinite but rather indefinite, he should be corrected by the grammarians. For the word 'indefinite' ought never to be applied to that which actually is, but always looks to a future possibility, signifying only something which is not yet determined and definite. Thus before God had decreed anything about the creation of the world (if ever he was not decreeing), the quantity of matter, the number of the stars, and all other things were indefinite; once the world was created, they were defined. Thus matter is indefinitely divisible, but is always divided either finitely or infinitely (Part I, article 26; Part II, article 34). Thus an indefinite line is one whose future length is still undetermined. And so an indefinite space is one whose future magnitude is not yet determined; for indeed that which actually is, is not to be defined, but either does or does not have boundaries and so is either finite or infinite. Nor may Descartes object that he takes space to be indefinite in relation to us; that is, we just do not know its limits and do not know positively that there are none (Part I, article 27). This is because although we are ignorant beings, God at least understands that there are no limits, not merely indefinitely but certainly and positively, and because although we negatively imagine it to transcend all limits, yet we positively and most certainly understand that it does so. But I see what Descartes feared, namely that if he should consider space infinite, it would perhaps become God because of the perfection of infinity. But by no means, for infinity is not perfection except when it is attributed to perfect things. Infinity of intellect, power, happiness, and so forth is the height of perfection; but infinity of ignorance, impotence, wretchedness, and so on is the height of imperfection; and infinity of extension is so far perfect as that which is extended.

3. The parts of space are motionless. If they moved, it would have to be said either that the motion of each part is a translation from the vicinity of other contiguous parts, as Descartes defined the motion of bodies, and it has been sufficiently demonstrated that this is absurd; or that it is a translation out of space into space, that is out of itself, unless perhaps it is said that two spaces everywhere coincide, a moving one and a motionless one. Moreover, the immobility of space will be best exemplified by duration. For just as the parts of duration are individuated by their order, so that (for example) if yesterday could change places with today and become the later of the two, it would lose its individuality and would no longer be yesterday, but today; so the parts of space are individuated by their positions, so that if any two could change their positions, they would change their individuality at the same time and each would be converted numerically into the other. The parts of duration and space are understood to be the same as they really are only because of their mutual order and position; nor do they have any principle of individuation apart from that order and position, which consequently cannot be altered.

4. Space is an affection of a being just as a being. No being exists or can exist which is not related to space in some way. God is everywhere, created minds are somewhere, and body is in the space that it occupies; and whatever is neither everywhere nor anywhere does not exist. And hence it follows that space is an emanative effect of the first existing being, for if any being whatsoever is posited, space is posited. And the same may be asserted of duration: for certainly both are affections or attributes of a being according to which the quantity of any thing's existence is individuated to the degree that the size of its presence and persistence is specified. So the quantity of the existence of God is eternal in relation to duration, and infinite in relation to the space in which he is present; and the quantity of the existence of a created thing is as great in relation to duration as the duration since the beginning of its existence, and in relation to the size of its presence, it is as great as the space in which it is present.

Moreover, lest anyone should for this reason imagine God to be like a body, extended and made of divisible parts, it should be known that spaces themselves are not actually divisible, and furthermore, that any being has a manner proper to itself of being present in spaces. For thus the relation of duration to space is very different from that of body to space. For we do not ascribe various durations to the different parts of space, but say that all endure simultaneously. The moment of duration is the same at Rome and at London, on the earth and on the stars, and throughout all the heavens. And just as we understand any moment of duration to be diffused throughout all spaces, according to its kind, without any concept of its parts, so it is no more contradictory that mind also, according to its kind, can be diffused through space without any concept of its parts.

5. The positions, distances, and local motions of bodies are to be referred to the parts of space. And this appears from the properties of space enumerated as 1 and 4 above, and will be more manifest if you conceive that there are vacuities scattered between the particles, or if you pay heed to what I have formerly said about motion. To this it may be further added that in space there is no force of any kind that might impede, assist, or in any way change the motions of bodies. And hence projectiles describe straight lines with a uniform motion unless they meet with an impediment from some other source. But more of this later.

6. Lastly, space is eternal in duration and immutable in nature because it is the emanative effect of an eternal and immutable being. If ever space had not existed, God at that time would have been nowhere; and hence he either created space later (where he was not present himself), or else, which is no less repugnant to reason, he created his own ubiquity. Next, although we can possibly imagine that there is nothing in space, yet we cannot think that space does not exist, just as we cannot think that there is no duration, even though it would be possible to suppose that nothing whatever endures. This is manifest from the spaces beyond the world, which we must suppose to exist (since we imagine the world to be finite), although they are neither revealed to us by God, nor known through perception, nor does their existence depend upon that of the spaces within the world. But it is usually believed that these spaces are nothing; yet indeed they are spaces. Although space may be empty of body, nevertheless it is not in itself a void; and something is there, because spaces are there, although nothing more than that. Yet in truth it must be acknowledged that space is no more space where the world exists, than where there is no world, unless perchance you would say that when God created the world in this space he at the same time created space in itself, or that if God should afterwards annihilate the world in this space, he would also annihilate the space in it. Whatever has more reality in one space than in another space belongs to body rather than to space; the same thing will appear more clearly if we lay aside that puerile and jejune prejudice according to which extension is inherent in bodies like an accident in a subject without which it cannot actually exist.

Now that extension has been described, it remains to give an explanation of the nature of body. Of this, however, the explanation must be more uncertain, for it does not exist necessarily but by divine will, because it is hardly given to us to know the limits of the divine power, that is to say, whether matter could be created in one way only, or whether there are several ways by which different beings similar to bodies could be produced. And although it scarcely seems credible that God could create beings similar to bodies which display all their actions and exhibit all their phenomena, and yet would not be bodies in essential and metaphysical constitution, as I have no clear and distinct perception[8]

[8] Here Newton has adopted Cartesian terminology familiar to his readers.

of this matter I should not dare to affirm the contrary, and hence I am reluctant to say positively what the nature of bodies is, but I would rather describe a certain kind of being similar in every way to bodies, and whose creation we cannot deny to be within the power of God, so that we can hardly say that it is not body.

Since each man is conscious that he can move his body at will, and believes further that other men enjoy the same power of similarly moving their bodies by thought alone, the free power of moving bodies at will can by no means be denied to God, whose faculty of thought is infinitely greater and more swift. And for the same reason it must be agreed that God, by the sole action of thinking and willing, can prevent a body from penetrating any space defined by certain limits.

If he should exercise this power, and cause some space projecting above the earth, like a mountain or any other body, to be impervious to bodies and thus stop or reflect light and all impinging things, it seems impossible that we should not consider this space really to be a body from the evidence of our senses (which constitute our sole judges in this matter); for it ought to be regarded as tangible on account of its impenetrability, and visible, opaque, and coloured on account of the reflection of light, and it will resonate when struck because the adjacent air will be moved by the blow.

Thus we may suppose that there are empty spaces scattered through the world, one of which, defined by certain limits, happens by divine power to be impervious to bodies, and by hypothesis it is manifest that this would resist the motions of bodies and perhaps reflect them, and assume all the properties of a corporeal particle, except that it will be regarded as motionless. If we should suppose that that impenetrability is not always maintained in the same part of space but can be transferred here and there according to certain laws, yet so that the quantity and shape of that impenetrable space are not changed, there will be no property of body which it does not possess. It would have shape, be tangible and mobile, and be capable of reflecting and being reflected, and constitute no less a part of the structure of things than any other corpuscle, and I do not see why it would not equally operate upon our minds and in turn be operated upon, because it would be nothing other than the effect of the divine mind produced in a definite quantity of space. For it is certain that God can stimulate our perception by means of his own will, and thence apply such power to the effects of his will.

In the same way, if several spaces of this kind should be impervious to bodies and to each other, they would all sustain the vicissitudes of corpuscles and exhibit the same phenomena. And so if all of this world were constituted out of these beings, it would hardly seem to be inhabited differently. And hence these beings will either be bodies, or very similar to bodies. If they are bodies, then we can define bodies as *determined quantities of extension which omnipresent God endows with certain conditions*. These conditions are: (1) that they be mobile, and therefore I did not say that they are numerical parts of space which are absolutely immobile, but only definite quantities which may be transferred from space to space; (2) that two of this kind cannot coincide anywhere, that is, that they may be impenetrable, and hence that oppositions obstruct their mutual motions and they are reflected in accord with certain laws; (3) that they can excite various perceptions of the senses and the imagination in created minds, and conversely be moved by them, which is not surprising since the description of their origin is founded on this.

Moreover, it will help to note the following points concerning the matters already explained.

1. That for the existence of these beings it is not necessary that we suppose some unintelligible substance to exist in which as subject there may be an inherent substantial form; extension and an act of the divine will are enough. Extension takes the place of the substantial subject in which the form of the body is conserved by the divine will; and that product of the divine will is the form or formal reason of the body denoting every dimension of space in which the body is to be produced.

2. These beings will not be less real than bodies, nor (I say) are they less able to be called substances. For whatever reality we believe to be present in bodies is conferred on account of their phenomena and sensible qualities. And hence we would judge these beings, since they can receive all qualities of this kind and can similarly exhibit all these phenomena, to be no less real, if they should exist in this manner. Nor will they be any less than substances, since they will likewise subsist and acquire accidents through God alone.

3. Between extension and its impressed form there is almost the same analogy that the Aristotelians posit between prime matter and substantial forms, namely when they say that the same matter is capable of assuming all forms, and borrows the denomination of numerical body from its

form.[9] For so I posit that any form may be transferred through any space, and everywhere denote the same body.

4. They differ, however, in that extension (since it [involves] 'what' and 'how constituted' and 'how much') has more reality than prime matter, and also in that it can be understood in the same way as the form that I assigned to bodies. For if there is any difficulty in this conception it is not in the form that God imparts to space, but in the manner by which he imparts it. But that is not to be regarded as a difficulty, since the same question arises with regard to the way we move our bodies, and nevertheless we do believe that we can move them. If that were known to us, by like reasoning we should also know how God can move bodies, and expel them from a certain space bounded in a given figure, and prevent the expelled bodies or any others from being able to enter it again, that is, cause that space to be impenetrable and assume the form of body.

5. Thus I have deduced a description of this corporeal nature from our faculty of moving our bodies, so that all the difficulties of the conception may at length be reduced to that; and further, so that God may appear (to our innermost consciousness) to have created the world solely by the act of will, just as we move our bodies by an act of will alone; and, moreover, so that I might show that the analogy between the divine faculties and our own may be shown to be greater than has formerly been perceived by philosophers. That we were created in God's image, holy writ testifies. And his image would shine more clearly in us if only he simulated in the faculties granted to us the power of creation in the same degree as his other attributes; nor is it an objection that we ourselves are created beings and so a share of this attribute could not have been equally granted to us. For if for this reason the power of creating minds is not delineated in any faculty of created mind, nevertheless created mind (since it is the image of God) is of a far more noble nature than body, so that perhaps it may eminently contain [body] in itself. Moreover, in moving bodies we create nothing, nor can we create anything, but we only simulate the power of creation. For we cannot

[9] A doctrine of so-called prime matter, according to which a type of "formless" matter would underlie various fundamental kinds of change that bodies, or elements, can undergo, is sometimes attributed to Aristotle. The attribution remains controversial: see *Physics* (190b and 193a), and *Generation and Corruption* (332a).

make any space impervious to bodies, but we only move bodies; and at that not any we choose, but only our own bodies, to which we are united not by our own will, but by divine constitution; nor can we move bodies in any way but only in accord with those laws which God has imposed on us. If anyone, however, prefers this our power to be called the finite and lowest level of the power which makes God the creator, this no more detracts from the divine power than it detracts from his intellect that intellect belongs to us in a finite degree, particularly since we do not move our bodies by a proper and independent power but by laws imposed on us by God. Rather, if anyone should think it possible that God may produce some intellectual creature so perfect that it could, by divine accord, in turn produce creatures of a lower order, this I submit does not detract from the divine power, it posits an infinitely greater power, by which creatures would be brought forth not only directly but by other intermediate creatures. And so some may perhaps prefer to posit a soul of the world created by God, upon which he imposes the law that definite spaces are endowed with corporeal properties, rather than to believe that this function is directly discharged by God. To be sure, the world should not be called the creature of that soul but of God alone, who creates it by constituting the soul of such a nature that the world necessarily emanates [from it]. But I do not see why God himself does not directly inform space with bodies, so long as we distinguish between the formal reason of bodies and the act of divine will. For it is contradictory that it [body] should be the act of willing or anything other than the effect which that act produces in space, which effect does not even differ less from that act than Cartesian space, or the substance of body according to the common concept; if only we suppose that they are created, that is, that they borrow existence from the will, or that they are beings of the divine reason.

Lastly, the usefulness of the idea of body that I have described is brought out by the fact that it clearly involves the principal truths of metaphysics and thoroughly confirms and explains them. For we cannot posit bodies of this kind without at the same time positing that God exists, and has created bodies in empty space out of nothing, and that they are beings distinct from created minds, but able to be united with minds. Say, if you can, which of the views, now common, elucidates any one of these truths or rather is not opposed to all of them, and leads to obscurity. If we say with Descartes that extension is body, do we not manifestly offer a path to atheism, both because extension is not created but has existed eternally, and because we have an idea of it without any relation to God, and so in some circumstances it would be possible for us to conceive of extension while supposing God not to exist? Nor is the distinction between mind and body in his philosophy intelligible, unless at the same time we say that mind has no extension at all, and so is not substantially present in any extension, that is, exists nowhere; which seems the same as if we were to say that it does not exist, or at least renders its union with body thoroughly unintelligible and impossible. Moreover, if the distinction of substances between thinking and extended is legitimate and complete, God does not eminently contain extension within himself and therefore cannot create it; but God and extension would be two separate, complete, absolute substances, and in the same sense. But on the contrary if extension is eminently contained in God, or the highest thinking being, certainly the idea of extension will be eminently contained within the idea of thinking, and hence the distinction between these ideas will not be such that both may fit the same created substance, that is, but that a body may think, and a thinking being be extended. But if we adopt the common idea (or rather lack of it) of body, according to which there resides in bodies some unintelligible reality that they call substance, in which all the qualities of the bodies are inherent, this (apart from its unintelligibility) is exposed to the same problems as the Cartesian view. Since it cannot be understood, it is impossible that its distinction from the substance of the mind should be understood. For the distinction drawn from substantial form or the attributes of substances is not enough: if bare substances do not have an essential difference, the same substantial forms or attributes can fit both, and render them by turns, if not at one and the same time, mind and body. And so if we do not understand that difference of substances deprived of attributes, we cannot knowingly assert that mind and body differ substantially. Or if they do differ, we cannot discover any basis for their union. Further, they attribute no less reality in concept (though less in words) to this corporeal substance regarded as being without qualities and forms, than they do to the substance of God, abstracted from his attributes. They conceive of both, when considered simply, in the same way; or rather they do not conceive of them, but confound them in some common apprehension of an unintelligible reality. And hence it is not surprising that atheists arise ascribing to corporeal substances that which solely belongs to the divine. Indeed, however we cast about we find almost no other reason for atheism than this notion of bodies having, as it were, a complete, absolute, and independent reality in themselves, such as almost all of us, through negligence, are accustomed to have in our minds from childhood (unless I am mistaken), so that it is only verbally that we call bodies created and dependent. And I believe that this prejudice explains why the same word, substance, is applied univocally in the schools to God and his creatures, and what philosophers, in forming the idea of body, cling to and ramble on about, when they try to form an independent idea of a thing dependent upon God. For certainly whatever cannot exist independently of God cannot be truly understood independently of the idea of God. God does not sustain his creatures any less than they sustain their accidents, so that created substance, whether you consider its degree of dependence or its degree of reality, is of an intermediate nature between God and accident. And hence the idea of it no less involves the concept of God, than the idea of accident involves the concept of created substance. And so it ought to embrace no other reality in itself than a derivative and incomplete reality. Thus the prejudice just mentioned must be laid aside, and substantial reality is to be ascribed to these kinds of attributes, which are real and intelligible things in themselves and do not need to be inherent in a subject, rather than to the subject which we cannot conceive as dependent, much less form any idea of it. And this we can manage without difficulty if (besides the idea of body expounded above) we reflect that we can conceive of space existing without any subject when we think of a vacuum. And hence some substantial reality fits this. But if, moreover, the mobility of the parts (as Descartes supposed) should be involved in the idea of vacuum, everyone would freely concede that it is corporeal substance. In the same way, if we should have an idea of that attribute or power by which God, through the action of his will alone, can create beings, we should readily conceive of that attribute as subsisting by itself without any substantial subject and [thus as] involving the rest of his attributes. But while we cannot form an idea of this attribute, nor even of our proper power by which we move our bodies, it would be rash to say what may be the substantial basis of mind.

So much for the nature of bodies, which in explicating I judge that I have sufficiently proved that such a creation as I have expounded is most clearly the work of God, and that if this world were not constituted from that creation, at least another very like it could be constituted. And since there is no difference between the materials as regards their properties and nature, but only in the method by which God created one and the other, the distinction between body and extension is certainly brought to light from this. For extension is eternal, infinite, uncreated, uniform throughout, not in the least mobile, nor capable of inducing change of motion in bodies or change of thought in the mind; whereas body is opposite in every respect, at least if God did not please to create it always and everywhere. For I should not dare to deny God that power. And if anyone thinks otherwise, let him say where he could have created prime matter, and whence the power of creating was granted to God. Or if there was no beginning to that power, but he had the same eternally that he has now, then he could have created from eternity. For it is the same to say that there never was in God an impotence to create, or that he always had the power to create and could have created, and that he could always create matter. In the same way, either a space may be assigned in which matter could not be created from the beginning, or it must be conceded that God could have created it everywhere.

Moreover, so that I may respond more concisely to Descartes's argument: let us abstract from body (as he demands) gravity, hardness, and all sensible qualities, so that nothing remains except what pertains to its essence. Will extension alone then remain? By no means. For we may also reject that faculty or power by which they [the qualities] stimulate the perceptions of thinking things. For since there is so great a distinction between the ideas of thought and of extension that it is not obvious that there is any basis of connection or relation [between them], except that which is caused by divine power, the above capacity of bodies can be rejected while preserving extension, but not while preserving their corporeal nature. Clearly the changes which can be induced in bodies by natural causes are only accidental and they do not denote that substance is really changed. But if any change is induced that transcends natural causes, it is more than accidental and radically affects the substance. And according to the sense of the demonstration, only those things are to be rejected which bodies can be deprived of, and made to lack, by the force of nature. But should anyone object that bodies not united to minds cannot directly arouse perceptions in minds, and that since there are bodies not united to minds, it follows that this power is not essential to them, it should be noticed that there is no suggestion here of an actual union, but only of a capacity of bodies by which they are capable of such a union through the forces of nature. From the fact that the parts of the brain, especially the more subtle ones to which the mind is united, are in a continual flux, new ones succeeding those which fly away, it is manifest that that capacity is in all bodies. And whether you consider divine action or corporeal nature, to remove this is no less than to remove that other faculty by which bodies are enabled to transfer mutual actions from one to another, that is, to reduce body into empty space.

However, as water offers less resistance to the motion of solid bodies passing through it than quicksilver does, and air much less than water, and aetherial spaces even less than air-filled ones, if we set aside altogether every force of resistance to the passage of bodies, we must also set aside the corporeal nature [of the medium] utterly and completely. In the same way, if the subtle matter were deprived of all forces of resistance to the motion of globules, I should no longer believe it to be subtle matter but a scattered vacuum. And so if there were any aerial or aetherial space of such a kind that it yielded without any resistance to the motions of comets or any other projectiles, I should believe that it was utterly empty. For it is impossible that a corporeal fluid should not impede the motion of bodies passing through it, assuming that (as I supposed before) it is not disposed to move at the same speed as the body.[10]

However, it is manifest that every force can be removed from space only if space and body differ from one another; and hence that each can be removed is not to be denied before it has been proved that they do not differ, lest an error be let in by begging the question.

But lest any doubt remain, it should be observed from what was said earlier that there are empty spaces in nature. For if the aether were a corporeal fluid entirely without vacuous pores, however subtle its parts are made by division, it would be as dense as any other fluid, and it

[10] Part II, Epistle 96 to Mersenne. (Editor's note: Newton probably means Descartes's letter to Mersenne of 9 January 1639, in which Descartes discusses the motion of bodies through various kinds of media. The original letter is available in *Oeuvres de Descartes*, ed. Charles Adam and Paul Tannery (Paris: Vrin, 1996), vol. II: 479–92, and is partially translated in *The Philosophical Writings of Descartes*, ed. John Cottingham et al. (Cambridge University Press, 1991), vol. III, 131–3. Thanks to Alan Gabbey for this reference.)

would yield to the motion of passing bodies with no less inertia; indeed with a much greater inertia if the projectile were porous, because then the aether would enter its internal pores, and encounter and resist not only the whole of its external surface, but also the surfaces of all the internal parts. Since the resistance of the aether is on the contrary so small when compared with the resistance of quicksilver as to be over ten or a hundred thousand times less, there is all the more reason for thinking that by far the largest part of the aetherial space is empty, scattered between the aetherial particles. The same may also be conjectured from the various gravities of these fluids, for the descent of heavy bodies and the oscillations of pendulums show that these are in proportion to their densities, or as the quantities of matter contained in equal spaces. But this is not the place to go into this.

Thus you see how fallacious and unsound this Cartesian argumentation is, for when the accidents of bodies have been rejected, there remains not extension alone, as he supposed, but also the capacities by which they can stimulate perceptions in the mind by means of various bodies. If we further reject these capacities and every power of moving, so that there only remains a precise conception of uniform space, will Descartes fabricate any vortices, any world, from this extension? Surely not, unless he first invokes God, who alone can generate new bodies in those spaces (or by restoring those capacities to the corporeal nature, as I explained above). And so in what has gone before I was correct in assigning the corporeal nature to the capacities already enumerated.

And so finally, since spaces are not the very bodies themselves, but are only the places in which bodies exist and move, I think that what I laid down concerning local motion is sufficiently confirmed. Nor do I see what more could be desired in this matter, unless perhaps I warn those for whom this is not satisfactory that by the space whose parts I have defined as places filled by bodies, they should understand the Cartesian generic space in which spaces regarded singularly, or Cartesian bodies, are moved, and so they will find hardly anything to object to in our definitions.

I have already digressed enough; let us return to the main theme.

Definition 5. Force is the causal principle of motion and rest. And it is either an external one that generates, destroys, or otherwise changes impressed motion in some body, or it is an internal principle by which existing motion or rest is conserved in a body, and by which any being endeavours to continue in its state and opposes resistance.

Definition 6 *Conatus* [endeavour] is resisted force, or force in so far as it is resisted.

Definition 7. *Impetus* is force in so far as it is impressed on a thing.

Definition 8. *Inertia* is the inner force of a body, lest its state should be easily changed by an external exciting force.

Definition 9. Pressure is the endeavour [conatus] of contiguous parts to penetrate into each other's dimensions. For if they could penetrate [each other] the pressure would cease. And pressure is only between contiguous parts, which in turn press upon others contiguous to them, until the pressure is transferred to the most remote parts of any body, whether hard, soft, or fluid. And upon this action is based the communication of motion by means of a point or surface of contact.

Definition 10. Gravity is the force in a body impelling it to descend. Here, however, by descent is not only meant a motion towards the centre of the earth, but also towards any point or region, or even from any point. In this way if the endeavour [conatus] of the aether gyrating about the sun to recede from its centre be taken for gravity, in receding from the sun the aether could be said to descend. And so by analogy, that plane should be called horizontal that is directly opposed to the direction of gravity or conatus. Moreover, the quantity of these powers, namely motion, force, conatus, impetus, inertia, pressure, and gravity, may be reckoned by a twofold account: that is, according to either its intension or extension.

Definition 11. The intension of any of the above mentioned powers is the degree of its quality.

Definition 12. Its extension is the quantity of space or time in which it operates.

Definition 13. Its absolute quantity is the product of its intension and its extension. So, if the quantity of the intension is 2, and the quantity of the extension 3, multiply the two together and you will have the absolute quantity 6.

Moreover, it will be helpful to illustrate these definitions via individual powers. And thus motion is either more intense or more remiss, as the space traversed in the same time is greater or less, for which reason a body is usually said to move more swiftly or more slowly. Again, motion is more or less extended as the body moved is greater or less, or as it is diffused through a larger or smaller body. And the absolute quantity of motion is composed of both the velocity and the magnitude of the moving body. So force, conatus, impetus, or inertia are more intense as they are greater in the same or an equivalent body: they are more extensive when the body is larger, and their absolute quantity arises from both. So the intension of pressure is proportional to the increase of pressure upon the surface area; its extension proportional to the surface pressed. And the absolute quantity results from the intension of the pressure and the quantity of the surface pressed. So, lastly, the intension of gravity is proportional to the specific gravity of the body; its extension is proportional to the size of the heavy body, and absolutely speaking the quantity of gravity is the product of the specific gravity and mass of the gravitating body. And whoever fails to distinguish these clearly, necessarily falls into many errors concerning the mechanical sciences.

In addition, the quantity of these powers may sometimes be reckoned according to the period of duration; for which reason there will be an absolute quantity which will be the product of intension, extension, and duration. In this way, if a body [of size] 2 is moved with a velocity 3 for a time 4, the whole motion will be $2 \times 3 \times 4$ or 24.[11]

Definition 14. Velocity is the intension of motion, slowness is remission.
Definition 15. Bodies are denser when their inertia is more intense, and rarer when it is more remiss.

The rest of the above mentioned powers have no names. It is, however, to be noted that if, with Descartes or Epicurus, we suppose rarefaction and condensation to be accomplished in the manner of relaxed or compressed sponges, that is, by the dilation and contraction of pores which are either filled with, or empty of, some very subtle matter, then we ought to estimate the size of the whole body from the quantity of both its parts and its pores as in Definition 15; so that one may consider inertia to be remitted by the increase of the pores and intensified by their diminution, as though the pores, which offer no inertial resistance to change, and whose mixtures with the truly corporeal parts give rise to all the various degrees of inertia, bear some ratio to the parts.

But in order that you may conceive of this composite body as a uniform one, suppose its parts to be infinitely divided and dispersed everywhere throughout the pores, so that in the whole composite body there is not the least particle of extension without an absolutely perfect mixture of infinitely divided parts and pores. Certainly such reasoning is suitable for contemplation by mathematicians; or if you prefer the manner of the peripatetics: things seem to be captured differently in physics.

[11] The original manuscript erroneously has "12" in place of "24."

Definition 16. An elastic body is one that can be condensed by the force of pressure or compressed within the limits of a narrower space; and a nonelastic body is one that cannot be condensed by that force.

Definition 17. A hard body is one whose parts do not yield to pressure.

Definition 18. A fluid body is one whose parts yield to an overwhelming pressure. Moreover, the pressures by which the fluid is driven in any direction whatsoever (whether these are exerted merely on the external surface, or on the internal parts by the action of gravity or any other cause), are said to be balanced when the fluid rests in equilibrium. This situation obtains if the pressure is exerted in some one direction and not towards all directions at once.

Definition 19. The limits defining the surface of the body (such as wood or glass) containing the fluid, or defining the surface of the external part of the same fluid containing some internal part, constitute the vessel of fluid.

In these definitions, however, I refer only to absolutely hard or fluid bodies, for one cannot reason mathematically concerning bodies that are partially so, on account of the innumerable figures, motions, and connections of the least particles. Thus I suppose that a fluid does not consist of hard particles, but that it is of such a kind that it has no small portion or particle which is not likewise fluid. And moreover, since the physical cause of fluidity is not to be examined here, I define the parts, not as being in motion among themselves, but only as capable of motion, that is, as being everywhere so divided one from another that, although they may be supposed to be in contact and at rest with respect to one another, yet they do not cohere as though stuck together, but can be moved separately by any impressed force and can change the state of rest as easily as the state of motion if they move relatively. Indeed, I suppose that the parts of hard bodies do not merely touch each other and remain at relative rest, but that they also so strongly and firmly cohere, and are so bound together – as it were by glue – that no one of them can be moved without all the rest being drawn along with it; or rather that a hard body is not made up of conglomerate parts, but is a single undivided and uniform body which preserves its shape most resolutely, whereas a fluid body is uniformly divided at all points.

And thus I have accommodated these definitions not to physical things but to mathematical reasoning, after the manner of the geometers who do not accommodate their definitions of figures to the irregularities of physical bodies. And just as the dimensions of physical bodies are best determined by their geometry – as with the dimension of a field by plane geometry, although a field is not a true plane; and the dimension of the earth by the doctrine of the sphere, even though the earth is not precisely spherical – so the properties of physical fluids and solids are best known from this mathematical doctrine, even though they are not perhaps absolutely nor uniformly fluid or solid as I have defined them here.

Axioms

1. From like postulates like consequences ensue.
2. Bodies in contact press one another equally.

Propositions on non-elastic fluids

Prop. 1. All the parts of a non-gravitating fluid, compressed with the same intension in all directions, press each other equally (or with equal intension).

Prop. 2. And compression does not cause motion between the parts.

Demonstration of both

Let us first suppose that the fluid is contained and uniformly compressed by the spherical boundary *AB* whose centre is *K* (Figure 3.1). Any small portion of it *CGEH* is bounded by the two spherical surfaces *CD* and *EF* described about the same centre *K* and by the conical surface *GKH* whose vertex is at *K*. And it is manifest that *CGEH* cannot in any way approach the centre *K* because all the matter between the spherical surfaces *CD* and *EF* would everywhere approach the same centre for the same reason,[12] and so would penetrate the volume of the fluid contained within the sphere *EF*.[13] Nor can *CGEH* recede in any direction towards the circumference *AB* because all that shell of fluid between *CD* and *EF* would similarly recede for the same reason,[14] and so would penetrate the volume of fluid between the spherical surfaces *AB* and *CD*.[15] Nor can it be squeezed out sideways, say towards *H*, since if we imagine another little section Hγ, terminated in every direction by the same spherical surfaces and a similar conical surface and contiguous to *GH* at *H*, this

[12] Axiom 1. [13] Contrary to the definition. [14] Axiom 1.
[15] Contrary to the definition.

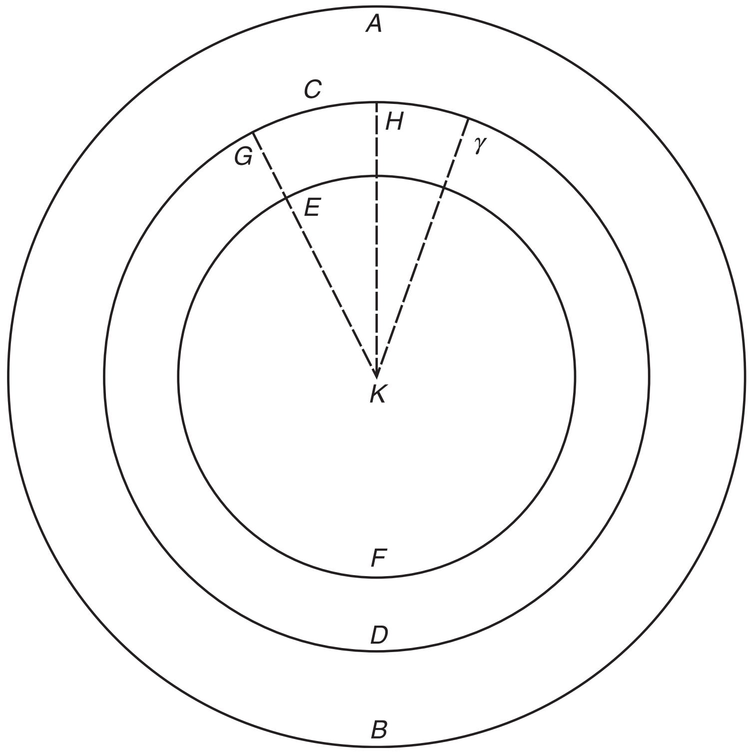

Figure 3.1

section Hγ may for the same reason be squeezed out towards H,[16] and so effect a penetration of volume by the mutual approach of contiguous parts.[17] And so it is that no portion of fluid CGEH can exceed its limits because of pressure. And hence all the parts remain in equilibrium. Which is what I wished to demonstrate first.

I saw also that all parts press each other equally, and with the same intension of pressure that the external surface is pressed. To show this, imagine that *PSQR* is a part of the said fluid *AB* contained by similar spherical segments *PRQ* and *PSQ*, and that its compression upon the internal surface *PSQ* is as great as that upon the external surface *PRQ* (Figure 3.2). For I have already shown that this part of the fluid remains in equilibrium, and so the effects of the pressures acting on both of its surfaces are equal, and hence the pressures are equal.[18,19]

[16] Axiom 1. [17] Contrary to the definition.
[18] Axiom. [19] Definition.

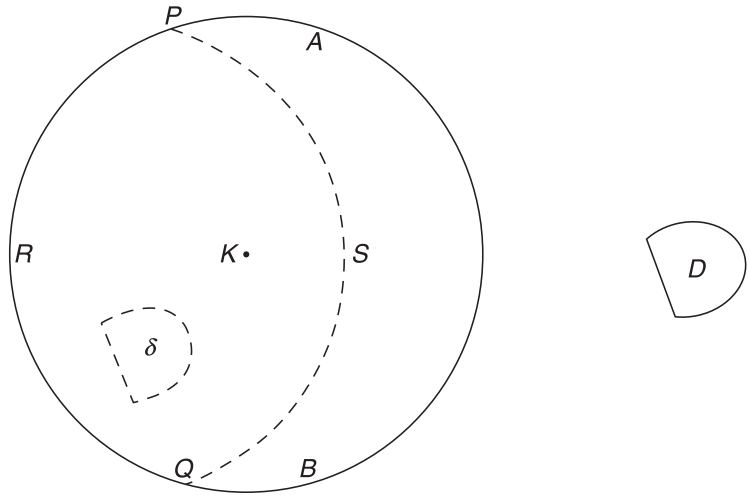

Figure 3.2

And thus since spherical surfaces such as *PSQ* can be described anywhere in the fluid *AB*, and can touch any other given surfaces in any points whatever, it follows that the intension of the pressure of the parts along the surfaces, wherever placed, is as great as the pressure on the external surface of the fluid. Which is the second point I wished to demonstrate.

Moreover, as the force of this argument is based on the equality of the surfaces *PRQ* and *PSQ*, lest it should seem that there is some disparity, in that one is within the fluid and the other is a segment of the external surface, it will help to imagine that the whole sphere *AB* is a part of an indefinitely larger volume of fluid, in which it is contained as within a vessel, and is everywhere compressed just as its part *PRQS* is pressed upon the surface *PSQ* by another part *PABQS*. For the method by which the sphere *AB* is compressed is of no significance, so long as its compression is supposed to be equal everywhere.

Now that these things have been demonstrated for a fluid sphere, I say lastly that all the parts of the fluid *D* (bounded in any manner at all, and compressed with the same intension in all directions) press each other equally and are not made to move relatively by the compression. For let *AB* be an indefinitely greater fluid sphere compressed with the same degree of intension; and let δ be some part of it equal and similar to *D*. From what has already been demonstrated it follows that this part δ is compressed with an equal intension in all directions and that the intension of the pressure is the same as that of the sphere *AB*, that is (by hypothesis) as that which compressed the fluid *D*. Thus the compression of the similar and equal fluids, *D* and δ, is equal; and hence the effects will be equal.[20] But all the parts of the sphere *AB*[21] and so of the fluid δ contained in it, press each other equally, and the pressure does not cause a relative motion of the parts. For which reason the same is true of the fluid *D*.[22] Q.E.D.

Corollary 1. The internal parts of a fluid press each other with the same intension as that by which the fluid is pressed on its external surface.

Corollary 2. If the intension of the pressure is not everywhere the same, the fluid does not remain in equilibrium. For since it stays in equilibrium because the pressure is everywhere uniform, if the pressure is anywhere increased, it will predominate there and cause the fluid to recede from that region.[23]

Corollary 3. If no motion is caused in a fluid by pressure, the intension of the pressure is everywhere the same. Since if it is not the same, motion will be caused by the predominant pressure.[24]

Corollary 4. A fluid presses on whatever bounds it with the same intension as the fluid is pressed by whatever bounds it, and vice versa. Since the parts of a fluid are certainly the bounds of contiguous parts and press each other with an equal intension, conceive the aforesaid fluid to be part of a greater fluid, or similar and equal to such a part, and similarly compressed, and the assertion will be evident.[25]

Corollary 5. A fluid everywhere presses all its bounds, if they are capable of withstanding the pressure applied, with that intension with which it is itself pressed in any place. For otherwise it would not be pressed everywhere with the same intension.[26] On which assumption it yields to the more intense pressure.[27] And so it will either be condensed or it will break through the bounds where the pressure is less.[28]

[20] Axiom. [21] According to what has been demonstrated.
[22] Axiom. [23] Definition. [24] Corollary 2. [25] Axiom. [26] Corollary 4.
[27] Corollary 2. [28] Contrary to the hypothesis.

Scholium

I have proposed all this about fluids, not as contained in hard and rigid vessels, but within soft and quite flexible bounds (say within the internal surface of a homogeneous exterior fluid), so that I might more clearly show that their equilibrium is caused only by an equal degree of pressure in all directions. But once a fluid is put into equilibrium by an equal pressure, it is equal whether you imagine it to be contained within rigid or yielding bounds.